

Jia-Rong Cheng

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PlusRon



Education

National Central University | Master degree - CSIE

Zhongli, Taiwan

M.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

GPA : 4.3 / 4.3

September 2023 - July 2025

National Yunlin University of Science and Technology | Bachelor degree - CSIE

Yunlin, Taiwan

B.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

GPA : 3.98 / 4.0 , 1ST PLACE OF THE DEPARTMENT

September 2019 - June 2023

Firmware & Linux Projects

Bare-metal ARM Cortex-M0 | SSD Controller Simulator PlusRon/Firmware-STM32-SSD-Emulator

PERSONAL PROJECT | SYSTEM-LEVEL FIRMWARE AND FTL ARCHITECTURE DESIGN

March 2026 - Present

- **Bootstrapping, Boot Sequence, and Custom HAL**

- Standalone GNU Toolchain built with ARM-GCC, Make, and OpenOCD for a CLI-based environment.
- Linker Scripts and Startup Code authored to manage memory layout, Vector Table configuration, and SCB-based auto-recovery.
- Custom HAL (STM32F072) engineered for precise peripheral control.

- **Interrupt-Driven Architecture**

- Implemented a Producer-Consumer model via Ring Buffer, integrating DMA and ISR to achieve non-blocking I/O.
- UART IDLE Detection and SysTick interrupts utilized for asynchronous event handling and system timing.

- **FTL (Flash Translation Layer) Management**

- L2P Mapping and Out-of-place Updates implemented to manage LBA-to-PBA address translation.
- Linked-list-based Garbage Collection (GC) developed for efficient page-pooling and stale data reclamation.

- **Communication Protocol & Reliability**

- NVMe-like Protocol Parser engineered with Checksum validation and Endianness handling.
- UART ORE detection and recovery were implemented to prevent Data Overflow and Race Conditions.
- Python-based Unit Testing developed to automate host-to-device protocol validation and command verification.

Linux Kernel | Custom Syscall for Page Table Traversal

PlusRon/linux_kernel

ACADEMIC PROJECT : LINUX OPERATING SYSTEM | SUPERVISED BY PROF. FU-HAU HSU, NCU

December 2025

- Custom System Call implemented for manual PGD-to-PTE traversal to retrieve physical addresses from virtual memory mappings of target processes.
- User-Kernel Data Exchange engineered to export memory mappings metadata for user-space diagnostics and dmesg verification.

Master's Thesis

SLC-E : Segment-Level Clustering and Extraction based Cross-Modal Video Retrieval and its Applications in Video Captioning

PlusRon/SLC-E-base-VC-DST

GRADUATE RESEARCH PROJECT | MULTIMODAL VIDEO UNDERSTANDING : CROSS-MODAL RETRIEVAL AND CAPTIONING

June 2025

- **Video Retrieval (SLC-E)**

- Integrated ViT with K-medoids++ (SLC) for feature compression and Cross-Modal Attention (CMSE) for segment alignment.
- Leveraged Contrastive Learning to optimize model training and boost video retrieval performance.

- **Video Captioning (DST)**
 - Dual-Stream Transformer implements a coupled architecture.
 - Utilized Auto-regressive decoding and Teacher Forcing to enhance generative text quality.
- **Benchmark Evaluation**
 - Achieved 6.4% gain in BLEU-4 (captioning) and 4.8% in R@1 (retrieval) on the MSR-VTT dataset.
 - Confirming high-precision semantic alignment across Retrieval and Captioning.

Work Experience

NCKU Tainan Industrial Senior High School | Intern Teacher

 **PlusRon/NCKU-ATIHS-intern**

DEPT. OF ELECTRONICS AND INFORMATION ENGINEERING

January 2026 - Present

- **Hardware Instrumentation & Debugging**
 - Proficient in Oscilloscopes, DMMs, and Signal Generators.
 - OPA / MOS / BJT / Digital Logic breadboard assemblies and perform systematic troubleshooting.
- **Circuit Simulation App**  **Flutter_app-Electronics_laboratory_project**
 - Developed a Flutter PWA based on Wasm and Skia, deployed via GitHub Actions CI/CD to Firebase.
 - Cache Busting and asynchronous Service Worker monitoring were implemented to ensure real-time version updates and resource consistency.
 - Simulates real-time circuit hardware behavior through high-performance software logic to enable interactive instructional demonstrations.

Skills

Programming & Scripting	C (Embedded) , Python (Testing/AI) , C++ , Shell Script , Makefile
Embedded Systems	STM32 (ARM Cortex-M0) , DMM , Oscilloscope , Linker Scripts , UART
Storage & FTL	L2P Mapping , Out-of-place Updates , Garbage Collection , NVMe-like Protocol
Linux Kernel	Custom Syscalls , PGD/PTE Traversal
Toolchains & DevOps	ARM GCC , GNU Make , OpenOCD , GDB , Linux CLI , Git , CI/CD , Firebase
AI & Machine Learning	PyTorch , Transformers , ViT , Contrastive Learning , Multimodal Learning